Full Test - Trigonometry

Time Allowed : 2 hour ____________________________ Maximum Marks : 200

Instructions

1. The test consists of two Parts.

Part I

2. All questions in Part I carry +5 marks for a correct

3. Question number 1 to 12 are single answer MCQ's . Each correct response carries +5 marks while for each incorrect response, 2 marks will be deducted.

4. Question number 13 to 19 are single answer MCQ's based on paragraphs. Each correct response carries +5 marks while for each incorrect response, 1 mark will be deducted.

5. Question number 20 is Assertion Reason type Question. A correct response carries +5 marks while 1 mark will be deducted for an incorrect response.

Part II

6. Question number 21 to 30 are MCQ's with none, one or more than one correct answers. Each correct response carries +8 marks while for each incorrect response, 3 marks will be deducted.

7. Question number 31 is a Matrix match type problem. For all correct rows +20 marks will be awarded , for one row incorrect, +10 marks will be awarded and for more than rows incorrect, no credit will be given.

Part I

1. If $\tan\theta = \dfrac{\sqrt{6}a}{2} - \dfrac{1}{2\sqrt{6}a}$, then $\sec\theta - \tan\theta$ is equal to

(A) $\sqrt{6}a, \dfrac{1}{\sqrt{6}a}$

(B) $-\sqrt{6}a, \dfrac{1}{\sqrt{6}a}$

(C) $\sqrt{6}a, -\dfrac{1}{\sqrt{6}a}$

(D) None of these

2. If $x = r\sin\alpha\sin\beta\cos\gamma$, $y = r\sin\alpha\sin\beta\sin\gamma$, $z = r\sin\alpha\cos\beta$, $w = r\cos\alpha$ then $x^2+y^2+z^2+w^2$ is independent of

A) r, α, β, γ

B) r, α, β

C) α, β, γ

D) None of these

3. The angles of a right angled triangle are in A.P. The ratio of the area of circumcircle of the triangle to the area of the triangle is

A) $\dfrac{2\pi}{\sqrt{3}}$

B) $\dfrac{\pi}{\sqrt{3}}$

C) $\dfrac{\pi}{3}$

D) None of these

4. The number of solutions of $8^{cosec^2 x} + 8^{-\cot^2 x} = \dfrac{33}{2}$, $0 \le x \le 2\pi$ is

A) 4

B) 6

C) 8

D) None of these

5. If $-\pi \le \theta \le \pi$ and r, θ satisfy $r\sin\theta = 3$ and $r = 3\,(2 + 3\sin\theta)$, then the number of ordered pairs (r, θ), which are solutions is/are

A) 0

B) 1

C) 2

D) None of these

6. If $0 \le x, y, z, t \le 2\pi$ and $\sin x + \dfrac{\sin y}{2} + \dfrac{\sin z}{3} = -t^2 + 2t - \dfrac{17}{6}$, then the value of $x + y + z$ is

A) $\dfrac{3\pi}{2}$

B) $\dfrac{5\pi}{2}$

C) $\dfrac{7\pi}{2}$

D) $\dfrac{9\pi}{2}$

7. $\tan\dfrac{\pi}{96}+2\tan\dfrac{\pi}{48}+4\tan\dfrac{\pi}{24}+8+8\sqrt{3}$ is equal to

A) $\cot\dfrac{\pi}{96}-8$

B) $\cot\dfrac{\pi}{96}$

C) 32

D) None of these

8. The number of distinct real solutions of

$\left(x^2+3\right)(cosecx+\cot x)-3x+\sqrt{3}\left[x\,(cosecx+\cot x)-\left(x^2+3\right)\right]=0$ on $-2\pi\le x\le 2\pi$ is

A) 1

B) 2

C) 4

D) None of these

9. In a $\triangle ABC$, with sides a,b,c and $\angle B=\dfrac{\pi}{6}$, $\angle C=\dfrac{\pi}{4}$. The area of the triangle is

A) $\dfrac{\left(\sqrt{3}-1\right)}{4}a^2$

B) $\dfrac{1}{2}a^2$

C) $\dfrac{\left(\sqrt{3}+1\right)}{4}a^2$

D) None of these

10. The area of $\triangle ABC$ is $(b+c)^2-a^2$. Then which of following statements is true

A) $\angle A$ is acute

B) $\angle A$ is right angle

C) $\angle A$ is obtuse

D) None of these

11. If $\dfrac{1}{\sqrt{2}}<x<1$, the number of solutions of the equations

$\tan^{-1}\left(\dfrac{1}{x}-1\right)+\tan^{-1}\left(\dfrac{1}{x}\right)+\tan^{-1}\left(\dfrac{1}{x}+1\right)=\tan^{-1}(3x)$ is

A) 0

B) 3

C) 4

D) None of these

12. The number of solutions of the equation $\sin^{-1}(1-x)-2\sin^{-1}(x)=\dfrac{\pi}{2}$ is/are

a) One

b) Two

c) Three

d) None of these

Comprehension 1: For a quadrilateral with sides a, b , c , d , semi-perimeter 's' and sum of any two opposite angles $= 2\alpha$, the area $\triangle$ is given by

$\triangle^2 = (s-a)(s-b)(s-c)(s-d) - abcd\cos^2\alpha$

Now, we take a special quadrilateral which can be inscribed in a circle C_1 and a circle C_2 is inscribed in it

13. The area of a special quadrilateral $\triangle$ is given by

A) $\sqrt{s(s-a)(s-b)(s-c)}$

B) $\dfrac{s^2}{3}$

C) $\sqrt{abcd}$

D) $\sqrt{3abcd}$

14. The radius of the inscribed circle C_2 is given by

A) $\dfrac{\triangle}{s}$

B) $\dfrac{2\triangle}{s}$

C) $\dfrac{3\triangle}{s}$

D) $\dfrac{4\triangle}{s}$

15. Cosine of the angle between the diagonals of the special quadrilateral is given by

A) $\pm\dfrac{ab-cd}{s^2}$

B) $\pm\dfrac{ac-bd}{\triangle}$

C) $\pm\dfrac{ab-cd}{ab+cd}$

D) $\pm\dfrac{ac-bd}{ac+bd}$

16. If B is the angle between sides a and b of the special quadrilateral then $tan\dfrac{B}{2}$ is given by

A) $\sqrt{\dfrac{ab}{cd}}$

B) $\sqrt{\dfrac{cd}{ab}}$

C) $\dfrac{ab}{cd}$

D) $\dfrac{cd}{ab}$

Comprehension 2: A function $f : \left(\frac{5\pi}{2}, \frac{7\pi}{2}\right) \to (-1, 1)$ is defined by $f(x) = \sin x$. Its inverse is denoted by $f^{-1}(x) = \sin^{-1} x$. Another function $g : (2\pi, 3\pi) \to (-1, 1)$ is defined by $g(x) = \cos x$. Its inverse is denoted by $g^{-1}(x) = \cos^{-1} x$. An onto function $h : (-1, 0) \to R_h$ is given by $h(x) = f^{-1}(x) - g^{-1}(x)$. Now answer the following questions.

17. $g^{-1}(-x)$ is given by

a) $\pi - g^{-1}(x)$
b) $5\pi - g^{-1}(x)$
c) $\frac{5\pi}{2} - g^{-1}(x)$
d) None of these

18. Set 'R_h' is

a) $\left\{\frac{\pi}{2}\right\}$
b) $\left(\frac{5\pi}{2}, 3\pi\right)$
c) $\left(0, \frac{\pi}{2}\right)$
d) None of these

19. If a function $v : (-\infty, 0] \to \left[3\pi, \frac{7\pi}{2}\right)$ is defined by $v(x) = f^{-1}\left(\frac{2^x - 2^{-x}}{2^x + 2^{-x}}\right)$, then $v(x)$ is

a) Injective Only
b) Surjective Only
c) Bijective
d) None of these.

20. The following question is an Assertion-Reason type question.The question contains two statements. Statement-1 (Assertion) and Statement-2(Reason). The question has five choices (A), (B) , (C) , (D) ,(E) out of which 'ONLY ONE' is correct.

(A) Statement-1 is true, Statement-2 is true; Statement 2 is the correct and complete explaination of Statement-1.

(B) Statement-1 is true, Statement-2 is true; Statement 2 is **NOT** the correct and complete explaination of Statement-1.

(C) Statement-1 is true, Statement-2 is false.

(D) Statement-1 is false, Statement-2 is true.

(E) Statement-1 is false, Statement-2 is false.

Statement 1: If the sides of a triangle are 3,4,5 then the altitudes of the triangle are $\frac{1}{3}, \frac{1}{4}, \frac{1}{5}$.

Statement 2: If the sides of a triangle are in A.P. then the altitudes of the triangle are in H.P.

Part II

21. If $\cos\theta = a$ for exactly one value of $\theta \epsilon \left[0, \frac{7\pi}{6}\right]$, then the value of 'a' can be

A) 0

B) $\frac{\sqrt{3}}{2}$

C) -1

D) $-\frac{1}{2}$

22. If $\sin^9 x + \cos^6 x = 1$ in the interval $-3\pi \le x \le 2\pi$, then which of these statements is/are correct

A) The number of roots of the equation is 7.

B) The sum of roots is -4π

C) The ratio of the number of roots on the left side of zero to the number of roots on the right side on the right side of zero, on the number line is $\frac{4}{3}$.

D) None of these

23. The equation $\cos^{-1} x = 3\cos^{-1} a$ has a solution for

A) all real values of a

B) all real values of a satisfying $a \le 1$

C) all real values of a satisfying $\frac{1}{2} \le a \le 1$

D) all real values of a satisfying $\frac{1}{\sqrt{2}} \le a \le \frac{\sqrt{3}}{2}$

24. If $\cos\alpha = \frac{1}{2}\left(x + \frac{1}{x}\right)$, $\sin\beta = \frac{1}{2}\left(y + \frac{1}{y}\right)$ where $x, y \epsilon \mathbb{R}$, then which of the following is/are correct.

A) $\cos(\alpha - \beta) = 0$

B) $|\sin(\alpha + \beta)| = 1$

C) $x = 1, y = 1$

D) None of these

25. If in a $\triangle ABC$, the incenter divides the median AD in the ratio $3:1$, then

A) $\triangle ABC$ is isosceles

B) $\triangle ABC$ is equilateral

C) $\cos A = \frac{8}{9}$

D) $\sin B = \sqrt{\frac{8}{9}}$

26. Two sides of a $\triangle$ are given by the roots of the equation $x^2 - 4x + 1 = 0$. The angle between them is given by $\sin\theta = \frac{4}{5}$. The third side may be

A) $2 - \sqrt{3}$

B) $\sqrt{\frac{76}{5}}$

C) $2 + \sqrt{3}$

D) $\frac{8}{\sqrt{5}}$

27. The equation $10\sin^2 x + 3\sin x - 4 = 0$ has exactly n roots in $\left[-\frac{n\pi}{4}, \frac{n\pi}{4}\right]$, $n\epsilon N$. Then $'n'$ can be

A) 4
B) 6
C) 7
D) None of these

28. For a $\triangle ABC$, the area of the triangle and the square of semi-perimeter are roots of the equation $x^2 - 42x + 216 = 0$. The quantity $\frac{1}{r_1} + \frac{1}{r_2} + \frac{1}{r_3}$ numerically equals(where r_1, r_2, r_3 are the ex-radii of $\triangle ABC$ and r is the inradius)

A) $\frac{1}{r}$
B) r
C) 1
D) None of these

29. The real values of $'x'$ for which $2\sin x\cos^3 x > \sin x \cos x$, where $-2\pi \leq x \leq 2\pi$ will belong to

A) $\left(0, \frac{\pi}{4}\right)$
B) $\left(-2\pi, -\frac{7\pi}{4}\right)$
C) $\left(\frac{3\pi}{2}, \frac{7\pi}{4}\right)$
D) None of these

30. If $\frac{1}{4\cos^4\theta + \sin^2 2\theta} = \frac{xy}{(x+y)^2}$, where $x\epsilon\mathbb{R}$, $y\epsilon\mathbb{R}$ is true , then the constraints on x and y include.

A) $x \neq y$
B) $x + y \neq 0$
C) $x \neq 0, y \neq 0$
D) $x = y, x \neq 0$

Trigonometry Test

PART - I

1	A	B	C	D
2	A	B	C	D
3	A	B	C	D
4	A	B	C	D
5	A	B	C	D
6	A	B	C	D
7	A	B	C	D
8	A	B	C	D
9	A	B	C	D
10	A	B	C	D
11	A	B	C	D
12	A	B	C	D

13	A	B	C	D
14	A	B	C	D
15	A	B	C	D
16	A	B	C	D
17	A	B	C	D
18	A	B	C	D
19	A	B	C	D

20	A	B	C	D	E

PART - II

21	A	B	C	D
22	A	B	C	D
23	A	B	C	D
24	A	B	C	D
25	A	B	C	D
26	A	B	C	D
27	A	B	C	D
28	A	B	C	D
29	A	B	C	D
30	A	B	C	D

31				
P	A	B	C	D
Q	A	B	C	D
R	A	B	C	D
S	A	B	C	D

Trigonometry Test

PART - I

1	A	**B**	C	D
2	A	B	**C**	D
3	**A**	B	C	D
4	**A**	B	C	D
5	A	B	C	**D**
6	A	B	C	**D**
7	**A**	B	C	D
8	A	B	**C**	D
9	**A**	B	C	D
10	A	B	**C**	D
11	**A**	B	C	D
12	**A**	B	C	D

13	A	B	**C**	D
14	**A**	B	C	D
15	A	B	C	**D**
16	A	**B**	C	D
17	A	**B**	C	D
18	**A**	B	C	D
19	A	B	**C**	D

20	A	B	C	**D**	E

PART - II

21	A	B	C	D
22	A	B	C	D
23	A	B	C	D
24	A	B	C	D
25	A	B	C	D
26	A	B	C	D
27	A	B	C	D
28	A	B	C	D
29	A	B	C	D
30	A	B	C	D

31				
P	A	B	C	D
Q	A	B	C	D
R	A	B	C	D
S	A	B	C	D